

Title : Interpolating the Stochastic Heat and Wave Equations with Time-independent Noise: Solvability and Exact Asymptotics.

General description of the content:

The paper aims at showing the well-posedness and asymptotic behavior of a solution to the (fractional) non-linear stochastic partial differential equation:

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2})u(t, x) = I_t^r \left[\sqrt{\theta}u(t, x)\dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2), \end{cases} \quad (0.1)$$

where $a \in (0, 2)$, $b \in (0, 2)$, $r \geq 0$, $\nu > 0$, and $\theta > 0$. Here W is a centered and only space dependent Gaussian process whose correlation function is given by

$$\gamma(x) = \prod_{i=1}^k |x_{(i)}|^{-\alpha_i}.$$

The authors prove that

- (1) (0.1) has a unique (global) solution $u(t, x)$ in $L^p(\Omega)$ for all $p \geq 2$ if

$$0 < \alpha < \min \left(\frac{a}{b}[2(b+r)-1], 2a, d \right),$$

where $\alpha = \sum_{i=1}^k \alpha_i$

- (2) Otherwise, if

$$r \in [0, 1/2] \quad \text{and} \quad 0 < \alpha = \frac{a}{b}[2(b+r)-1] \leq d,$$

(0.1) is locally well-posed for a time $t \in (0, T_p)$, where

$$T_p := \frac{\nu^{\alpha/a}}{2\theta(p-1)\mathcal{M}_a^{(2a-\alpha)/a}},$$

$$\mathcal{M}_a(\gamma, \theta) = \sup_{g \in \mathcal{F}_a} \left\{ \left(\int \int_{\mathbb{R}^{2d}} g^2(x)g^2(y)\gamma(x+y)dxdy \right)^{1/2} - \frac{\theta}{2}\varepsilon_a(g, g) \right\},$$

$$\varepsilon_a(g, g) := (2\pi)^{-d} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi,$$

and

$$\mathcal{F}_a = \{f \in L^2(\mathbb{R}^d) : \|f\|_{L^2(\mathbb{R}^d)} = 1, \varepsilon_a(f, f) < \infty\}.$$

Moreover they proved that the solution satisfies the following asymptotic behavior

$$\begin{aligned} & \lim_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \\ &= \frac{1}{2} \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \left(\theta \nu^{-\alpha/1} \mathcal{M}_a^{\frac{2a-\alpha}{a}} \right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right), \end{aligned}$$

where

$$\beta = \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1} \quad \text{and} \quad t_p = (p-1)^{1-1/\beta} t.$$

General evaluation:

Definitely, I think that the paper is interesting and very well-written. Since studies for fractional stochastic PDEs are very popular these days, the results fit to a recent trend in SPDEs and seems to be interesting enough to attract attention of mathematicians in this field. Moreover since the authors provided many interesting examples, I could understand how general their results are. I think that this result naturally initiates lots of interesting subsequent works handling more general random noises and non-linearity. In total, I recommend this paper to be published in Stochastics and Partial Differential Equations: Analysis and Computations.

Here is the list of comments which seems to be easily handled.

- p2] +15. a nonnegative and nonnegative measure on $\mathbb{R}^d$.

I think that “a nonnegative and nonnegative definite measure on $\mathbb{R}^d$ ” is right.

- p4] -1. $\mathcal{M}_{a,d}(f)$.

Since the constant $\mathcal{M}_{a,d}(f)$ plays a very important role in the main results, I think that it is better to give more details about $\mathcal{M}_{a,d}$ and recall them in Section 3.2.

- p13] -9. $\mathcal{A}_d$.

Is $\mathcal{A}_d$ is right? or $\mathcal{A}_a$?

- p15] -8. $u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t))$.

Since this Wiener chaos expansion of the solution is the most important part of the theorems, I think that more details or appropriate references for the derivation should be given.

- p21] -7. $\sigma(a, d, \alpha)$.

Here $\sigma(a, d, \alpha)$ is used but $\sigma(d, \alpha)$ is used in the proof of Lemma 3.10 and Lemma 3.11. I recommend to unify them (choose $\sigma(a, d, \alpha)$ or $\sigma(d, \alpha)$).

- p29] -11. are are

One of “are” must be deleted.